$$k = \frac{\text{Rate}}{[M]^n}$$

$$k = \frac{7.5 \times 10^{-3}}{(0.05)^2} = 3 \text{ mol}^{-1}\text{Ls}^{-1}$$

12. Solution

$$\text{rate} = k[x]^{\left(\frac{3}{2}\right)}[y]^{\left(\frac{1}{2}\right)}$$

$$\text{overall order} = \left(\frac{3}{2} + \frac{1}{2}\right) = 2$$

i.e., second order reaction.

Since the rate expression does not contain the concentration of z , the reaction is zero order with respect to z.

15. Solution:

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$$

$$k = \frac{2.303}{1\,\text{min}} \log \frac{[0.08]}{[0.04]}$$

$$k = 2.303 \log 2$$

$$k = 2.303 \times 0.3010$$

$$k = 0.6932\,\text{min}^{-1}$$

$$k = \left(\frac{0.6932}{60}\right) s^{-1}$$

$$k = 1.153 \times 10^{-2}\ s^{-1}$$

19. Solution

$$k = A\, e^{-\left(\frac{E_a}{RT}\right)}$$

$$k = 1.6 \times 10^{13} \text{s}^{-1} e^{-\left(\frac{200 \times 103 \text{ J mol}^{-1}}{8.314 \text{JK}^{-1}\text{mol}^{-1} \times 600 \text{K}}\right)}$$

$$k = 1.6 \times 10^{13} \text{s}^{-1}\ e^{-(40.1)}$$

$$k = 1.6 \times 10^{13} \text{s}^{-1} \times 3.8 \times 10^{-18}$$

$$k = 6.21 \times 10^{-5} \text{s}^{-1}$$

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